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Me or Just Like Me?

The Role of Virtual Try-On and Physical Appearance in Apparel M-retailing.

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Abstract: This research investigated a crucial topic in apparel m-commerce: product presentation and the extent to which it should exhibit human visuals corresponding to consumers' appearance. A mobile application with virtual try-on (VTO) tool based on augmented reality was compared to a traditional m-commerce interface showing models with physical features not, partially, or completely similar to the consumers'. A theoretical framework based on the Technology Acceptance Model was adopted to explain the impact of the application and the mediating role of perceived hedonic value (enjoyment) and utilitarian value (convenience, ease of use, and usefulness) on attitude toward the shopping technology and purchase intention. An online experiment (415 respondents) and a qualitative study (49 respondents) showed that the VTO tool was less enjoyable than traditional m-commerce interfaces and less convenient and useful than pictures of models with physical features similar to those of consumers. Implications for academics and managers are discussed.

Key words: *Apparel m-retailing; Augmented reality; Human model; Technology Acceptance Model (TAM); Virtual Try On.*

1. Introduction

Apparel e-retailing is rapidly expanding worldwide[1], and mobile technology is a key factor in retail sector growth, accounting for 58.9% of online retail sales[2]. Marketers are increasingly interested in using mobile technology to meet consumer demand (Kumar et al., 2018). Doherty and Ellis-Chadwick (2010) state that mobile applications are the future of the retailing commerce. Nevertheless, mobile or "m-" commerce (as used in the following development) is an emerging field (Kumar et al., 2018). Researchers should investigate how to deliver utilitarian and hedonic value to consumers and create rich shopping experiences with retailing mobile applications (Lee and Kim, 2018). Consumers cannot physically examine and try on garments presented on e-retailing websites and m-commerce applications. Therefore, the success of apparel e-retailers depends to a large extent on convincing product presentation that helps consumers better visualize garments (e.g., Beck and Crié, 2018; Kim and Lennon, 2008; Yoo and Kim, 2014).

In traditional product presentation in apparel e-retailing, human models display garments. Virtual try-on (VTO) tools have been developed to help consumers evaluate garments by delivering information comparable to direct experience with products (Cho and Schwarz, 2012; Kim and Forsythe, 2008). Offline and online stores are constructing VTO tools based on augmented reality (AR) (e.g., Beck and Crié, 2018; Javornik, 2016a) to achieve multiple benefits (e.g., Rese et al., 2017; Yim et al, 2017). AR superimposes computer-generated virtual objects on real environments and allows real-time interactions (Javornik, 2016a), thus enriching a user's experience of reality (Poushneh and Vasquez-Parraga, 2017a). AR-based VTO tools have been developed and studied primarily in the context of jewelry, glasses, and make-up (e.g., Cho and Schwarz, 2012; Hilken et al., 2017; Rese et al., 2017; Yim et al., 2017). Yet, AR research must enlarge the contexts of use that are being investigated (Hilken et al., 2018), and apparel is a

relevant and important setting (e.g., Baytar et al., 2016; Beck and Crié, 2018). AR-based VTO tools represent expensive investments for apparel e-retailers, but it is not yet established whether they are more efficient than traditional pictures of human models wearing garments. The question of the effectiveness of AR as compared to traditional ways of product presentation is crucial for online retailers and has to be investigated thoroughly in order to bring strong evidence (Xu et al., 2018).

Early types of VTO utilized avatar- or photo-based try-on, while the most recent technologies rely on AR, which superimposes a three-dimensional (3D) virtual object on a moving reflection of the consumer in real time, acting like a digital mirror (Javornik et al., 2016). Some studies have assessed the relative effectiveness of different types of VTO (e.g., Javornik, 2016b; Merle et al., 2012), while others have compared VTO with traditional e-catalog presentations (e.g., Beck and Crié, 2018; Rese et al., 2017; Yim et al., 2017) or VTO with actual try-on (e.g., Baytar et al., 2016). However, except for Shin and Baytar (2014) in the context of an avatar-based VTO, existing research has not compared pictures of human models with an AR-based VTO tool and their respective impacts on the consumer's behavior.

Surprisingly, prior research in online shopping has given relatively little attention to the impacts of presenting human models in pictures (e.g., Berg, 2015). Research has compared flat presentations with human models (Kim et al., 2009; Yoo and Lennon, 2014) and studied the impact of the human model's body size on various consumers' responses (e.g., St-Onge et al., 2017). Interestingly, research also showed that idealized human models can sometimes trigger better consumer responses than an avatar-based VTO (Shin and Baytar, 2014). Nevertheless, while the advertising literature has demonstrated that ads are more effective when the physical features of human models match those of consumers (e.g., Chevalier and Lichtlé, 2012), little research in the e- and m-commerce literature has addressed the impacts of the fit between consumers and presenting human models.

With AR-based VTOs, female consumers can see themselves wearing garments. Augmentation technology impacts the consumer's sense of self (Scholz and Duffy, 2018) and affects how consumers perceive themselves (Javornik and Pizzetti, 2017). AR applications offers multiple benefits to the consumer, such as a better understanding of their own self and a possibility to take risks and discover aspects of their inner self while trying products they would not have tried otherwise (Scholz and Duffy, 2018). However, the "augmented self" is a "hybridization" between the consumer's actual body and virtual elements that create a *modified* perception of the self (e.g., Javornik and Pizzetti, 2017; Scholz and Duffy, 2018). The positive impact of AR-based VTO on consumer evaluations and behavioral responses is not obvious and might depend on certain conditions such as the female consumer's perception of the augmented image of herself that she sees on the screen (e.g., Cho and Schwarz, 2012). Thus, AR-based VTO tools might be less effective at triggering purchase intention than presentations of garments on human models with similar appearances because human models could serve an aspirational role (e.g., Jones, 2010) and reflect possible better versions of consumers' selves than their own appearance represented by augmented technology. The aim of this research, therefore, is to investigate whether AR-based VTO tools are more effective than pictures of human models in the m-commerce setting.

Drawing on the Technology Acceptance Model (TAM) (Davis, 1989) this research compares an m-commerce interface with an AR-based VTO tool and an m-commerce interface with traditional pictures of human models with physical features similar to consumers. A current research stream in the AR literature investigates whether VTO has utilitarian or hedonic value (Javornik, 2016a). Studies have shown that both aspects are important in the purchase process (e.g., Hilken et al., 2017; Merle et al., 2012), while others have pointed out the possible prevalence of one aspect over the other (e.g., Javornik et al., 2016; Kim and Forsythe, 2008) and their varying importance (Rese et al., 2017). Therefore, the paper studies the impacts of the m-commerce

interfaces on perceived hedonic value (perceived enjoyment) and utilitarian value (perceived convenience in examining the product, perceived ease of use, and perceived usefulness) and assesses how these values mediate the impact of the m-commerce interfaces on the attitude toward the shopping technology and purchase intention (Childers et al., 2002; Davis, 1989). Indeed, attitude toward the shopping technology (e.g., Chen et al., 2002; Javornik, 2016b; Yim et al., 2017) and purchase intention (e.g., Poushneh and Vasquez-Parraga, 2017a; Yim et al., 2017) are two relevant and important final variables in online retailing (Childers et al., 2001) and AR research (Hilken et al., 2018). Furthermore, following recommendations from the literature (e.g., Borau and Bonnefon, 2017; Keh et al., 2016), this study explores the impacts of the fit between female consumers and human models in pictures with regard to the important factors in self-identification: body size and ethnicity.

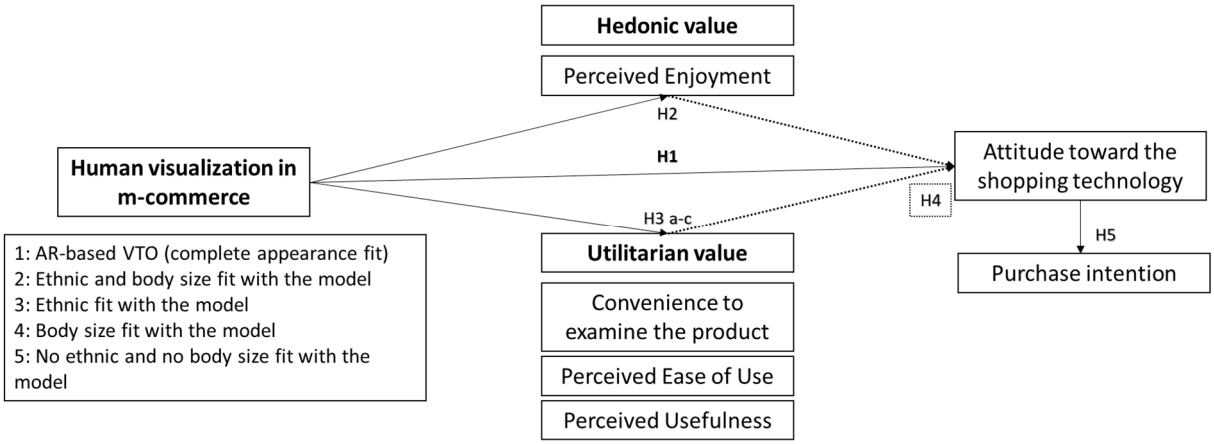
An experimental study with 415 usable responses and a qualitative study with 49 respondents offer interesting results. First, the results show that the AR-based VTO application is less enjoyable than any traditional m-commerce interface with pictures of human models. Perceived enjoyment mediates the effect of the m-commerce interface on the attitude toward the shopping technology and partly explains the final purchase intention. Second, a traditional m-commerce interface with pictures of human models having physical features similar to consumers' (i.e., same ethnicity and body size) is considered to be both more convenient and more useful than an m-commerce interface with an AR-based VTO tool. The no-fit traditional m-commerce interface is still significantly less convenient and useful than an AR-based VTO tool. Finally, attitudes toward the shopping technology are directly affected by its hedonic value (perceived enjoyment) and utilitarian value (perceived convenience of examining the product, perceived ease of use, and perceived usefulness), while the effect of the hedonic value prevails over utilitarian values.

This research offers three important contributions. First, it contributes to the growing field of online product presentation in digital commerce (e.g., Beck and Crié, 2018; Yoo and Kim, 2014) and enriches the scant literature on the impacts of human models in pictures (e.g., Berg, 2015; Kim et al., 2009). Second, this research studies the relative impacts of a traditional human model presentation compared to an AR-based VTO tool (e.g., Beck and Crié, 2018; Yim et al., 2017) and adds to the debate on the utilitarian and hedonic roles of such presentation formats (e.g., Javornik, 2016a; Rese et al., 2017). Third, this study contributes to the discussion in the marketing literature on the need to match the presenting human model with the target consumer (e.g., Chevalier and Lichtlé, 2012; Keh et al., 2016) and brings new insights from the m-retailing context. Finally, this research also provides possible explanations for VTO tools' lack of success and puts forward applicable solutions for apparel retailers.

2. Research Model and Hypotheses Development

We drew on the TAM to compare the relative impacts of an AR-based VTO mobile application and traditional m-commerce interfaces featuring human models with physical features similar to those of consumers on attitudes toward the shopping technology and purchase intention. The fit between female consumers and human models in pictures could be judged in terms of ethnicity, body size, and combination of these features due to the interactions of physical attributes in consumer perceptions (Keh et al., 2016). Following the literature in consumer research, we addressed Asian, Black, and White ethnicities (races) (e.g., D'Alessandro and Chitty, 2011; Jones, 2010; Keh et al., 2016). We investigated how the m-commerce interface affects both hedonic value (i.e., perceived enjoyment) and utilitarian value (i.e., perceived convenience of examining the product, perceived ease of use, and perceived usefulness) (Childers et al., 2001). Indeed, the TAM, which has been widely applied to investigate user acceptance of

new information technology, focuses on the roles of ease of use, usefulness, and enjoyment in predicting attitudes toward new technology (Davis et al., 1992). Thus, this framework is relevant to investigating the roles of both utilitarian and hedonic value in predicting attitudes toward VTO and AR applications (Kim and Forsythe, 2008; Merle et al., 2012; Rese et al., 2017). The following conceptual model is proposed (Fig. 1) and the hypotheses development is discussed hereon:



Note: the appearance fit with the m-commerce human model was established based on the similar ethnicity (Asian, Black, or White) and similar body size (thin or curvy)

Figure 1. Conceptual model.

2.1. The Impact of Online Product Presentation

2.1.1. Human Model Presentation

Online apparel stores traditionally use human models to present garments and facilitate product evaluation (Kim et al., 2009; St-Onge et al., 2017). Human models are proven to be more effective than flat presentations (Kim et al., 2009; Yoo and Lennon, 2014). Indeed, the presence of a human model on product presentation allows to activate the effect of self-referencing (Lee et al., 2002). Self-referencing occurs when consumer identifies something characteristic to oneself in the product offer (Lee et al., 2002). For instance, a human model might remind a female consumer of herself. Existing research suggests that activating the self-referencing by exposing the consumer to something/someone that she/he can relate to heightens recall of

information and can generate more favorable attitudes (Meyers-Levy and Peracchio 1996; Krishnamurthy and Sujan 1999).

A central question in advertising literature has been the fit between the physical features of human models and consumers (e.g., Chevalier and Lichtlé, 2012). The question of the match between presented human models and consumers is underpinned by self-congruence theory: people may identify with human models based on their self-concept (Sirgy, 1982). When consumers are exposed to visual stimuli with human models, they engage in a social comparison process (Buunk and Dijkstra, 2011; Pounders et al., 2017; Yu, 2014) and compare their own selves with the commercial images (Keh et al., 2016; Richins, 1991). According to Sirgy (1982), the congruity between the perceived self-image and product presentation stimulates consumers' positive attitudes and purchase intentions. Rhee and Johnson (2012) found support that the perceived similarity of apparel brand to the actual self has the strongest effect on the clothing brand preferences. Several studies have demonstrated that advertisements are also more effective when the physical features of human models match those of consumers because consumers identify with human models in pictures according to their perceived similarities (Chevalier and Lichtlé, 2012).

Human models possess multiple physical attributes that can influence consumer behavior. Several attributes, such as human models' ethnicity (e.g., D'Alessandro and Chitty, 2011), body size (e.g., Janssen and Paas, 2014; Pounders et al., 2017), gender (e.g., Berg, 2015), and age (e.g., Chevalier and Lichtlé, 2012), have been studied in advertising research. The empirical studies have shown that skin color is the most salient of all bodily appearances (Brunsma and Rockquemore, 2001; Grabe and Hyde, 2006). Ethnic identity is a crucial part of social identity in general, which serves to identify oneself as similar to ones and different from the others, and impacts on the creation of one's perceived self (Jenkins, 1994). In fact, ethnicity, gender, and body-size form a triad of consistent self-identification elements (e.g., Powlishta et

al., 1994). Furthermore, ethnicity and body-size perceptions are closely related: female consumers tend to compare themselves with human models and evaluate their perceived similarities based on ethnicity and body size (e.g., D'Alessandro and Chitty, 2011; Keh et al., 2016) and the body perception is very different among different ethnicities (e.g., Crago and Shisslak, 2003).

Existing studies have argued that consumers react more favorably to human models of the same ethnicity (e.g., Sierra et al., 2009). Results on the impact of the human model's body size are sometimes contradictory. While some studies in advertising find that the human model's body size has no main effect on consumers' responses (e.g., Roberts and Roberts, 2015), others show a clear relationship and demonstrate that moderately thin models are more effective than overly thin models in general (e.g., Janssen and Paas, 2014; Yu, 2014). The human model's body size on the website picture impacts the model's perceived attractiveness and this relationship is moderated by the congruence with the model (St-Onge et al., 2017). Surprisingly, human models with ideal thin bodies sometimes trigger better consumer responses than avatar-based VTO as they rise less concerns about the garment's size and fit than a 3D avatar built with the consumer's own measurement (Shin and Baytar, 2014). However, most importantly, consumers' reactions to thin and large models depend on their *own body mass index* (BMI). Women with low BMIs react positively to thin models (e.g., Smeesters et al., 2010), while human models with larger body sizes might not be good options for consumers with high BMIs because they can cause repulsion and make advertising less effective (e.g., Borau and Bonnefon, 2017).

Traditional advertising offers indirect experiences with products, whereas online commerce is aimed at creating direct experiences (Cho and Schwarz, 2012; Li et al., 2001) to engage consumers in thorough evaluation of products (Kim and Lennon, 2008). While the role of human models may be different in online commerce and advertising settings (Yoo and Kim,

2014), we expect the impact of the fit between human models and consumers to be positive in m-commerce context as well.

2.1.2. VTO Technology in online retailing

As the online retailing sector has grown, the lack of direct experience with products has been identified as a major issue, and virtual tools have been developed to create more direct and interactive experiences with products (Cho and Schwarz, 2012). To reduce the gap between physical stores and webstores, engineers have developed sales tools such as virtual fitting rooms to help consumers evaluate garments' fit before buying them (Beck and Crié, 2018). Early VTO technologies consisted of avatar-based simulations in which consumers created 3D virtual models corresponding to their own physical features (e.g., Merle et al., 2012; Shim and Lee, 2011; Shin and Baytar, 2014). The virtual models were only proxies of consumers, though, so photo-based try-on was developed to allow consumers to upload their own photos to try products (e.g., Cho and Schwarz, 2012). These tools did not allow consumers to try on products in real time in a moving environment. The recent development of AR, though, has opened new avenues.

AR superimposes computer-generated virtual objects on real environments and allows real-time interactions (Javornik, 2016a). As confirmed by Hilken et al. (2017), AR-based augmentation relies simultaneously on providing simulated control over the product (i.e., being able to perform natural movements to adjust the product) and environmental embedding (i.e., projecting the product on a personally relevant context such as the consumer's body or environment). AR has emerged as a highly relevant interactive marketing technology, especially in the retailing field to help consumers virtually try on products in real time (Javornik, 2016a). AR applications act as digital mirrors (Javornik et al., 2016) that enrich consumers' shopping experiences in multiple ways and allow consumers to visualize garments on

themselves (Hilken et al., 2017). As such, AR-based VTO applications allow to visualize the garment on a human presentation maximally similar to the customer's appearance. In the light of the above our baseline hypothesis is:

> **H1**. The more the consumer's physical features correspond to the human visualized in the m-commerce product presentation, the better is her attitude toward the shopping technology.

2.2. The Mediation Role of Utilitarian and Hedonic Values in Shopping Technology Evaluation

Visual product presentation aids on websites, such as pictures and videos, can create joyful shopping experiences and generate positive attitudes toward using websites (Jiang and Benbasat, 2007). VTO tools have specifically proven to generate hedonic value for consumers (e.g., Merle et al., 2012), especially female shoppers (Kim and Forsythe, 2008). Interestingly, perceived similarity or self-congruity with 3D avatar models (in body size and personal features such as ethnicity) positively influences perceived hedonic value, which then increases consumers' responses such as purchase intention (Merle et al., 2012). Thus, perceived similarity with presenting human models might create more enjoyment than non-fit with human models, which in its turn will lead to a better attitude toward the shopping technology (Childers et al., 2001).

Recent results have shown that AR-based VTO tools are specifically able to generate fun and offer playful experiences for consumers (Javornik et al., 2016). AR-based product presentations even result in higher enjoyment than traditional website presentations (Yim et al., 2017). The consumer's perceived hedonic value generated by the AR experience positively impacts consumers' responses such as purchase intention or return intention (e.g., Hilken et al., 2017; Javornik et al., 2016). Moreover, in the case of VTO, perceived enjoyment has been

proven to positively impact attitudes toward using it (Kim and Forsythe, 2008), especially when the VTO tool is based on AR (Yim et al., 2017). Therefore, based on these results, we propose a mediating effect of perceived enjoyment between the human visualized on the m-commerce product presentation and the consumer's attitude toward the shopping technology:

H2. The relationship between the consumer's physical resemblance with the human visualized in the m-commerce product presentation and the consumer's attitude toward the shopping technology is mediated by the perceived enjoyment.

According to previous results, simple visuals of products and product visualization tools (e.g., product rotation) on webstores generate utilitarian value by giving consumers information (e.g., Kim and Lennon, 2008; Park et al., 2008). VTO tools also have proven utilitarian benefits for consumers (e.g., Merle et al., 2012). Interestingly, the more an avatar-based VTO tool allows personalization and adaptation to consumers' body size and personal features such as ethnicity, the more its utilitarian value increases (Merle et al., 2012). Indeed, perceived self-congruence with virtual models' body size helps consumers evaluate garment fit and reduces perceived risk (Shim and Lee, 2011).

With AR-based VTO, consumers can see themselves precisely *as if they were wearing products*, which might facilitate their evaluation of products' fit (Javornik et al., 2016). Perceived as credible (Javornik et al., 2016), AR-based tools have been found to generate greater usefulness than traditional web-based product presentations (Yim et al., 2017). Therefore, it is expected that AR-based VTO tools deliver higher utilitarian value than traditional pictures of products worn by human models, even if the human models' physical features fit the consumers'. The AR experience increases utilitarian value, which in its turn increases the consumer's responses such as purchase or return intention (e.g., Hilken et al., 2017; Javornik et al., 2016). Furthermore, the perceived usefulness of the VTO tool positively

affects attitudes toward using it (Kim and Forsythe, 2008). The perceived ease of use and usefulness of an e-retailing AR application also positively influences attitudes toward using it (Rese et al., 2017; Yim et al., 2017). Thus, we draw on these results to propose a mediating effect of utilitarian values between the human visualized in the m-commerce product presentation and the consumer's attitude toward the shopping technology:

> **H3**. The relationship between the consumer's physical resemblance with the human visualized in the m-commerce product presentation and the consumer's attitude toward the shopping technology is mediated by the perceived convenience of examining the product (H3a), ease of use (H3b), and usefulness (H3c).

Several studies have demonstrated the utilitarian and hedonic value of online shopping technologies in general (e.g., Childers et al., 2001) and VTO tools in particular (e.g., Dacko, 2016; Javornik et al., 2016; Kim and Forsythe, 2008; Merle et al., 2012; Yim et al., 2017). In the case of AR-based applications, positive behavioral responses such as purchase intention or intention to return arise from increased utilitarian and hedonic value perceptions of the AR experience (e.g., Hilken et al., 2017; Javornik et al., 2016). Research on VTO tools and AR applications should address the relative importance of utilitarian and hedonic value in predicting and explaining consumers' responses (Javornik, 2016a). Some results in AR research point out that consumers value utilitarian benefits the most (e.g., Dacko, 2016). However, in some cases, the hedonic value of a VTO tool is a stronger cause for adoption than its utilitarian value (e.g., Kim and Forsythe, 2008). Moreover, some results suggest that the hedonic value of an AR-based VTO tool is a stronger predictor of consumer's behavioral intentions than the utilitarian value (Javornik et al., 2016). The relative importance of utilitarian and hedonic values seems in fact to vary among different types of AR applications and technologies (Rese et al., 2017). The intimate nature and context of use of an AR application can make its hedonic value prevail over its transactional

and utilitarian functions, in particular when the application helps the consumers to understand and define their own self (Scholz and Duffy, 2018). The prevalence of hedonic value over utilitarian value has been suggested in the context of an AR-based make-up application (Javornik et al.; 2016; Scholz and Duffy, 2018). Like make-up, clothes are personal goods that strongly relate to one's personality and self. Thus, in the context of apparel online retailing, we expect the hedonic value to have a stronger mediating effect on the consumer's attitude toward the shopping technology than the utilitarian value:

> **H4**. The hedonic value (perceived enjoyment) of the m-commerce product presentation has a stronger mediating effect on the attitude toward the shopping technology than utilitarian values (perceived convenience of examining the product, ease of use, and usefulness).

2.3. Impact of Attitude Toward the Shopping Technology on Purchase Intention

According to existing research, purchase intention is one of the consumer responses that matters the most for e-retailers and academics (e.g., Kim et al., 2009, Yim et al., 2017; Yoo and Kim, 2014). The relationship between the attitude to the online technology and the consequent purchase intention is well documented (e.g., Ranganathan and Jha, 2007; Van der Heijden et al., 2003). As posits the encyclopedic Technology Acceptance Model (Davis, 1989), the use of the technology depends on the formed attitude toward it. Positive attitude toward an online shopping technology is a corner-stone element in predicting further purchasing behavior (Kim, 2012). While attitude to the m-commerce relies on the perceived convenience, enjoyment, ease of use, and usefulness of the technology, it directly contributes to the behavioral intentions of purchasing on an m-commerce platform (Lu and Su, 2009). Previous works on online product presentation (e.g., Jiang and Benbasat, 2007; Park et al., 2008) and VTO (e.g., Merle et al., 2012; Yim et al., 2017) have also shown that consumers' attitudes toward the shopping interface positively impacts

their purchase intention. Indeed, the better consumers evaluate and the more they intend to use an m-commerce tool, the higher is the probability of making a purchase via this tool (e.g., Kim and Forsythe, 2008; Rese et al., 2017). Based on these results, the following hypothesis is proposed:

> **H5**. The e-consumer's attitude toward the shopping technology positively influences purchase intention.

3. Study 1

3.1. Data and Method

An experimental study compared an AR-based VTO application with an m-commerce interface displaying pictures of human models wearing garments. The respondents were randomly assigned to the VTO and the m-commerce conditions. To manipulate the resemblance of the shown m-commerce human model to the consumer in terms of body size and ethnicity, the m-commerce manipulation employed a 2 (human model's body size: thin vs. plus size) × 3 (human model's ethnicity: Asian vs. Black vs. White) between-subjects experimental design. The respondents were recruited on Amazon Mechanical Turk (AMT), a crowdsourcing platform which receives significant support for academic experimental studies (Rand, 2012). The questionnaire was addressed to women and was visible only to female master respondents from the US. Master qualification is granted by the AMT for a consistent success in performing a wide range of tasks on the AMT platform and thus assures high quality responses.

3.1.1. Experimental Design

The respondents first evaluated their actual physical appearance based on the schematic images showing female figures of different ethnicities and body size. We proposed multiple pictorial icons for the three studied ethnicities (i.e., with different hairstyles and eye shapes) and pictorial

icons for every type of body size from underweight to obese. These data were used to identify the objective fit of the consumers with the human models in the experimental condition (i.e., similar ethnicity and body size; different ethnicity but similar body size; similar ethnicity but similar body size; no similarities in ethnicity and body size). Next, they reported their actual height and weight. The actual weight and height were used to calculate the Body Mass Index (weight in kilograms divided by square height in meters) for an additional control of the self-declared body size. Respondents who significantly misjudged their body size according to the BMI or who communicated incomplete and unreasonable measures (e.g., 50 cm tall and weight of 170 kg) were eliminated. Further on, the respondents evaluated their attitudes toward and experience with m-commerce and retailing shopping.

The fit with a human visual on the product presentation was manipulated in the context of an m-commerce shopping interface. Respondents were shown a full-page screenshot of a mock mobile commerce application. The visual pictured one of the six human models (Asian curvy, Asian thin, Black curvy, Black thin, White curvy, or White thin) wearing a black dress. The m-commerce interface was enhanced with the design similar to many m-commerce applications on the market, description of the dress ("An elegant black dress that will suit to any body shape and to any occasion"), option to choose the size and add the dress to the cart. The AR-based VTO was introduced with a video showing an application that allowed uploading personal photos for virtual fittings and using either frontal or back cameras in an augmented VTO. The video featured a black and a dark blue dress. Examples of the visuals used in the experiment are provided in Appendix 1. After being exposed to the manipulation, the respondents were asked to describe the human model (i.e., her body size and ethnicity) and measure the perceived similarity with the human model. These measures were used to check the effectiveness of the manipulation. Further on, respondents answered about the perceived shopping experience, the shown garment, their attitude toward the shopping technology, and

their purchase intention. An attention check was also included (Appendix 2).

A pre-test was conducted with 22 respondents: the respondents were asked to look at all the visuals one by one and describe them as detailed as possible. This showed that the human models' physical features (i.e., Asian, Black, or Caucasian ethnicity and thin or plus-size body size) were recognized according to the intended manipulations. Further on, the pre-test respondents looked at all the visuals together and compared the dresses: all the dresses in the 6 m-commerce conditions were evaluated as relatively equal. Finally, the respondents were asked to watch the video on the AR-based VTO application and confirmed that the application allowed to visualize a product on their photo and to maximize the physical fit between themselves and the human visualization in an m-commerce product presentation.

3.1.2. Participants

A total of 654 US female respondents were randomly assigned to one of the seven experimental manipulations in an online survey. The respondents were recruited in two periods: Autumn 2017 (traditional m-commerce conditions) and Winter 2018 (AR-based VTO condition). An automated randomization mechanism ensured equal distribution of the respondents among the different experimental cells. The respondents were filtered to ensure they had sufficient experience with shopping technology and involvement in online retailing shopping (we selected only the respondents who evaluated their online shopping experience and involvement as higher than 4.5 on a 7-point scale). As the experimental conditions address m-commerce, we used only the questionnaires of the respondents who accessed the questionnaire via a mobile device. We also excluded respondents who could not identify with any of the ethnicities included in the experimentation (i.e., Asian, Black, or White) or who had a mixed ethnicity, as the actual fit between the model and the consumer could not be established. A total of 415 respondents remained in the final analysis. The sample thus consists of women from 18 to 72 years old, with

the majority 26–35 years old (53.6%); 35.2% of respondents were Asian, 23.8% were Black, and 41% had White ethnic background (Table 1). More than a half of respondents had thin to normal body complexion (60.7%), while 39.3% were overweight or obese. Most respondents held a bachelor's (42.3%) or a master's (21.8%) degree.

The final sample was divided into five studied conditions based on the fit between the consumer and the visualized human model on the product presentation: (1) AR-based VTO app allowing visualization of one-self and hence almost perfect fit of the consumer and human visualization on the product presentation; (2) ethnicity and body size fit on the m-commerce app (complete fit with the human model); (2) ethnicity fit only; (4) body size fit only; and (5) no fit condition (i.e., human model of different body size and ethnicity). The sample sizes bigger than the suggested 50 respondents per condition (VanVoorhis and Morgan, 2007) (i.e., 79–89 respondents per experimental cell) and the distribution allowed for acceptable comparability (Perreault and Darden, 1975) and data reliability (Simmons et al., 2011). We found no significant differences among the cells in age or ethnicity (all F-tests, $p > .50$), indicating that the samples are comparable (Table 2). The manipulation checks were satisfactory: human models were evaluated according to the measured physical features (i.e., thin or curvy; Asian, Black, or White). Furthermore, the perceived similarity with the m-commerce human model were significantly different for complete, partial, and no fit conditions ($p < .001$): the VTO and full fit conditions were perceived to offer a human visualization "similar to how I actually look" than partial and no fit conditions, while partial fit provided significantly more similarity than no fit condition.

Table 1
Sample Description

Age	Percentage	Ethnicity	Percentage	BMI	Percentage	Education	Percentage
18-25	25.5%	Asian	35.2%	Underweight (<18.5)	9.3%	High-school	21.4%
26-35	49.2%	Black	23.8%	Normal weight (18.5-24.9)	51.4%	Bachelor's degree	42.3%
36-45	15.2%	White	41%	Overweight (25-29.9)	21.5%	Master's degree	21.8%
46-55	5.3%			Obesity (>30)	17.8%	Other	14.5%
56-65	2.9%						
65-75	1.9%						

Note: respondents were female only from the United States of America (including the outlying islands)

Table 2
Experimental Cells Description

		Full Fit	Ethnicity Fit	Body size Fit	No Fit	AR-based VTO
Age	18-25	22	22	20	19	23
	26-35	41	40	40	41	42
	36-45	12	12	13	12	14
	46-55	4	5	4	4	5
	56-65	2	2	3	2	3
	65-75	2	2	1	1	2
Ethnicity	Asian	28	30	27	31	30
	Black	20	19	18	21	21
	White	34	33	32	35	36
BMI	Underweight	9	6	7	7	10
	Normal	43	45	42	44	39
	Overweight	16	18	16	19	20
	Obese	13	15	16	16	14
Education	High-school	16	17	17	18	21
	Bachelor's	37	33	38	34	34
	Master's	19	20	17	16	18
	Other	11	12	13	11	13
Total		83	83	81	79	89

3.1.3. Measures

The respondents completed the questionnaire using a 7-point Likert scale (from 1 "strongly disagree" to 7 "strongly agree") adopted from existing research (Appendix 2). Purchase intention was measured with a 3-item scale from Chandran and Morwitz (2005). The items on attitudes toward shopping technology were adapted from Chen et al. (2002), and the items on perceived ease of use and usefulness were taken from Davis (1989). The items on perceived enjoyment and convenience of examining the product were adapted from Childers et al. (2001).

Principal components analysis with varimax rotation was used to test the scale items for validity and reliability (Appendix 3). All the standardized factor loadings were greater than 0.841 and therefore deemed to be acceptable (Marsh and Hau, 1999). The internal consistency was satisfactory, with Cronbach's alpha of more than 0.822 and composite reliability of more than 0.919 (Chin, 1998). Convergent validity was established by examining the factor loadings and average variance extracted (AVE). All the items loaded significantly on their posited underlying constructs, and all the AVE scores were more than 0.80 (Fornell and Larcker, 1981). For discriminant validity, we checked that the intercorrelations between the latent factors did not include unity (Anderson and Gerbing, 1988) and that the AVE of each latent construct was greater than the squared correlations between any set of two constructs (Fornell and Larcker, 1981) (Appendices 3 and 4).

3.2. Results

Multivariate analysis of variance (MANOVA) was performed using SPSS (Hair et al., 2010) to assess the impacts of the stimuli on a number of dependent variables. MANCOVA was conducted to "establish the statistical significance of group differences in whole profiles of predictor variables and to reduce the number of predictors for a classification analysis" (Wang et al., 2001, p. 323). This analysis method was deemed to be the most appropriate for the categorical nature of the independent variables (Grace et al., 2015) and the expected interrelations between the dependent variables studied (Warne, 2014). MANOVA tested the direct effects of fit with the m-commerce human model on the respondents' attitudes and behavioral intentions (Table 3). Perceived convenience of examining the product on an m-commerce interface, perceived enjoyment and usefulness, as well as consequent purchase intention are significantly different for the experimental conditions. To establish the between-group difference the post-hoc tests were run using Tukey method (Abdi and Williams, 2010)

(Appendix 5). The ethnicity and age of the respondents were included as control variables, but were insignificant and, therefore, are not reported. The mean and standard deviation of measured variables for every experimental condition are presented in Table 4.

Table 3
Direct Effects

Independent variable	Dependent variables	F-value	Significance	Partial η^2
Human visualization in an m-commerce product presentation	P. Convenience of examining the product	33.67	.005	0.111
	Perceived Enjoyment	39.08	.000	0.149
	Perceived ease of use		.328	0.010
	Perceived usefulness	3.59	.017	0.020
	Attitudes toward the shopping technology	1.58	.121	0.014
	Purchase intention	3.33	.036	0.030

P.: Perceived

Table 4
Mean and standard deviation of the variables

Independent Variables	VTO Application[a] Mean (Std. Dev.)	M-commerce with Full Model Fit[b] Mean (Std. Dev.)	M-commerce with Ethnicity Model Fit[c] Mean (Std. Dev.)	M-commerce with Body Size Model Fit[d] Mean (Std. Dev.)	M-commerce with No Model Fit[e] Mean (Std. Dev.)
Dependent Variables					
P. convenience of examining the product**	4.81 (1.44)[b,e]	5.37 (1.11)[a,e]	4.92 (1.54)	4.93 (1.64)	4.41 (1.71)[a,b]
Perceived enjoyment***	3.28 (1.42) [b,c,d,e]	5.41 (1.21)[a]	5.5 (0.90)[a]	5.17 (1.54)[a]	5.17 (1.56)[a]
Perceived ease of use	5.26 (1.47)	5.55 (1.28)	5.61 (0.95)	5.64 (1.29)	5.53 (1.24)
Perceived usefulness*	4.85 (1.50)[c]	5.22 (1.13)	5.62 (0.96)[a]	5.34 (1.40)	5.10 (1.41)
Attitudes toward the shopping technology	5.12 (1.52)	5.73 (1.18)	5.60 (1.08)	5.26 (1.45)	5.38 (1.45)
Purchase intention*	4.84 (1.58)	5.42 (1.44)[e]	5.28 (1.15)	4.90 (1.86)	4.65 (1.91)[b]

* p-value < .050, ** p-value < .010, *** p-value < .001
Letters indicate significant differences with experimental conditions.
Std. Dev.: standard deviation
P.: Perceived

The results showed that the AR-based VTO application was less convenient for

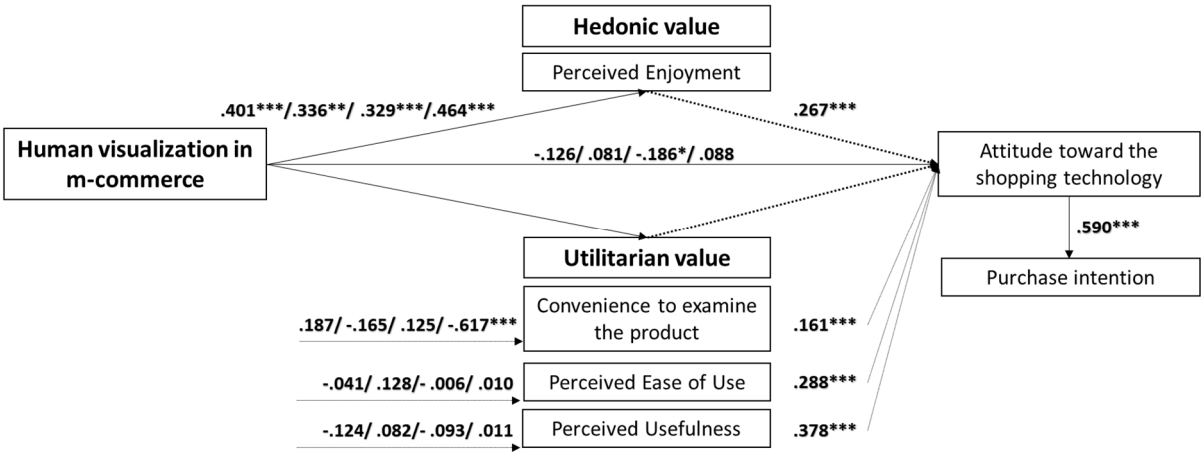
examining garments than the traditional m-commerce interface with full fit with human model (i.e., same ethnicity and body size) but was more convenient than the traditional m-commerce interface with no fit with the human model (M_{VT}=4.81 vs. $M_{FullFit}$=5.37 vs. M_{NoFit}=4.41, p-value<.010) (Table 3). Regarding perceived enjoyment from using the shopping interface, the VTO application scored lower than all types of traditional m-commerce interfaces (M_{VT}=3.28 vs. $M_{FullFit}$=5.41 vs. $M_{EthnicFit}$=5.50 vs. $M_{SizeFit}$=5.17 vs. M_{NoFit}=4.41, p-value < .001). The m-commerce interfaces did not differ significantly in perceived ease of use, although traditional m-commerce interfaces with ethnic fit were perceived as more useful than the AR-based VTO application (M_{VT}=4.85 vs. $M_{EthnicFit}$=5.62, p-value < .050). Finally, the traditional m-commerce interface with full fit with the human model led to significantly higher purchase intention than the traditional m-commerce interface with no fit ($M_{FullFit}$=5.42 vs. M_{NoFit}=4.65, p-value < .050). As no direct effect of m-commerce interface on the attitude toward the shopping technology is found, we cannot validate H1.

According to the supplementary linear regressions (Table 5), attitude toward the shopping technology was directly affected by the perceived convenience of examining products (B=.151, p-value<.001), enjoyment (B=.230, p-value<.001), ease of use (B=.309, p-value<.001), and usefulness (B=.369, p-value<.001). The employed variables explained up to 60.1% of intention to use the shopping interface. Finally, the purchase intention is directly and positively impacted by the attitude toward the shopping technology (B=.710, p-value<.001, R^2=49.6%) [H5].

Table 5
Multiple Linear Regressions

Explained variables	Attitude toward the Shopping Technology
Independent variables	
Perceived convenience to examine the product	.151***
Perceived enjoyment	.230***
Perceived ease of use	.309***
Perceived usefulness	.369***
R^2	60.1%

The direct effect is not necessary to establish a mediation (Zhao et al., 2010). To prove that the m-commerce interface largely explained the final consumer attitude, the mediation effect of the perceived hedonic and utilitarian values of the m-commerce interface was formally examined. Mediation analysis with a multicategorical independent variable was performed using the process macro in SPSS (Hayes and Preacher, 2014): model 6 with multiple mediators. Mediation analyses were conducted using the bootstrapping method developed by Hayes (2013), with each analysis employing N = 5,000 bootstrapped samples. Figure 2 presents the mediation results. We found significant differences between the AR-based VTO application and the traditional m-commerce interface in perceived convenience and enjoyment. A traditional m-commerce interface with no fit with the human model was less convenient than the AR-based VTO tool and thus led to a lower attitude toward the shopping technology (indirect effect: B_{NoFit}=-.128, CI 95% (-0.142; -.048)). All the traditional m-commerce interfaces led to higher perceived enjoyment than the AR-based VTO tool and consequently to a better attitude toward the shopping technology (indirect effects: $B_{FullFit}$=.124, CI 95% (.117; .420); $B_{EthnicFit}$=.089, CI 95% (.041; .253); $B_{SizeFit}$=.080, CI 95% (.060; .690); B_{NoFit}=.107, CI 95% (.047; .272)). Perceived ease of use and usefulness of the m-commerce interface did not mediate the effect of the m-commerce interface on the attitude toward the shopping technology. Indirect effects are presented in the Table 6.



Note: presentations of the results Full Fit/Ethnic Fit/ Body Size Fit/ No Fit as compared to AR-based VTO; * p-value < .100, ** p-value < .010, *** p-value < .001

Fig. 2. Mediation by hedonic and utilitarian values.

Table 6
Relative Indirect Effects

	AR-based VTO vs Full Fit	AR-based VTO vs Ethnic Fit	AR-based VTO vs Body Size Fit	AR-based VTO vs No Fit
Mediator	Coefficient (Confidence Interval at 95% Level of Confidence)*			
P. Enjoyment	.124 (.117; .420)	.089 (.041; .253)	.080 (.060; .690)	.107 (.047; .272)
P. Convenience to Examine the Product	-.148 (-.264; .056)	-.208 (-.337; .106)	-.154 (-.265; .066)	-.128 (-.142; -.048)
P. Ease of Use	-.003 (-.135; .132)	.023 (-.102; .159)	.002 (-.124; .129)	-.039 (-.167; .085)
P. Usefulness	-.113 (-.215; .009)	-.061 (-.171; .053)	-.106 (-.219, .005)	-.114 (-.223; .010)

P.: Perceived
* the end points of a 95% bootstrap confidence interval are the values of the 2.5th and 97.5th percentiles of the distribution, which shows that with 95% of probability the coefficient is within the interval.

Therefore, we found that the effects of the m-commerce interface on purchase intention and attitudes toward the shopping technology were partly explained by the perceived enjoyment (H2) and perceived convenience of examining the product using the technology (H3a). Further analysis of pairwise contrasts of indirect effects allowed us to compare the mediation effects of the utilitarian and hedonic values. The results are presented in Table 7. We found that the effects of the m-commerce interface on the attitude toward the shopping technology and consequently on the purchase decision are explained to a larger extent by the hedonic (perceived enjoyment) rather than utilitarian values (perceived convenience of examining the product, ease of use, and

usefulness) [H4]. The results also showed that perceived convenience of examining the garment is more important than perceived usefulness and ease of use of the shopping technology, while perceived usefulness is more important than the ease of use of the shopping technology.

Table 7
Comparison of Indirect Effects

Indirect effect contrast definitions		**Indirect effect**
Indirect effect 1	Indirect effect 2	Indirect effect 1 – Indirect effect 2
P. Enjoyment	P. Convenience	.108*
P. Enjoyment	P. Ease of use	.076*
P. Enjoyment	P. Usefulness	.068*
P. Convenience	P. Ease of use	.039*
P. Convenience	P. Usefulness	.031*
P. Ease of use	P. Usefulness	-.008*

* p-value < .050
P.: Perceived

4. Study 2: Exploration of the VTO Application

4.1. Method

To understand the more negative attitudes toward the AR-based VTO application than traditional m-commerce interfaces with pictures featuring human models, we conducted a qualitative study. The study mobilized two consumer groups: undergraduate students (20-25 years old) and adults (26-65 years old). The studies were carried out in Winter 2018 and Spring 2019 respectively.

First, a group of 48 undergraduate students were asked to download the app shown as the teaser video in the first experiment and to use it to shop for a garment for a university gala dinner. The chosen application was identified as one of the most advanced AR-based VTO applications available on the platform of mobile applications. According to the Google Play Store, the application counts more than 100.000 downloads. Although the application is positioned as an AR experience, not all of the features advertised in the video are available and the extent of the AR is limited. Thus, the users can download a personal photo and fit flat

visualization of the garments on the personalized photo. The pose in the photo can vary and the garment adjusts to the pose identified by the application. The choice of the background (e.g., indoors or outdoors) and the movement of the realistic background provides the AR part of the VTO experience.

After a 20-minute in-class shopping experience, the participants described their experiences in writing, based on the same dimensions as used in Study 1: perceived convenience of examining the fit of garments, enjoyment from the shopping experience, ease of use, and usefulness of the VTO application. At the end of the questionnaire, the respondents were asked to comment on their overall attitudes toward the app, their intention to use it for further shopping activities, and whether they found a garment they were willing to purchase for the gala dinner. To generate deeper insights into the shopping experience using the AR-based VTO application, the students were organized into groups of 8 for focus group discussions lasting 30-35 minutes. To improve the comparability of the results from studies 1 and 2, we excluded all the responses from male students and only analyzed the responses from 34 female students.

As a second step, we contacted 15 adult female consumers and asked them to download and experience the same m-commerce application. Following their independent experience with the application we carried out individual interviews, lasting on average 30 minutes. The sample of the qualitative study is described in the Table 8 and the results are presented below.

Table 8
Sample Description

Age	Respondents	Ethnicity	Respondents	Country of origin	Respondents	Education	Respondents
20-25	34	Asian	2	US	10	Bachelor's degree	36
26-35	8	Black	1	Mexico	7	Master's degree	7

36-45	3	White	46	France	26	Ph.D.	4
56-65	4			China	2	Other	2
				Russia	4		

Note: all 49 the respondents were female

4.2. Results

The results of the qualitative study (Table 9) showed that the main drawback of the application was the sophistication of the technology employed. The idea of being able to try on garments on oneself was very exciting, but the application had a number of technical and structural limitations. Indeed, the AR tool can enhance consumer experience, but only when it is of the top quality (Poushneh and Vasquez-Parraga, 2017b). However, the application allowed only a flat representation of the dress and the body, which did not permit evaluating a realistic fit of the garment on the body. Thus, despite a moderate level of interactivity and control of the tool (the respondents could change the photo and the background, try on many different clothes, and adjust the clothes to their shape), the overall AR experience is of low quality and, therefore, insufficient to engage the users (Poushneh and Vasquez-Parraga, 2017b).

Indeed, since the tool captures only the body of the user and superimposes it on a realistic, but recreated background, the environmental embedding is rather problematic (Hilken et al., 2017). Moreover, despite multiple function of fitting the type and the size of the garments, the clothes pictured on a small human representation does not allow the complete physical control over the virtual garment (Hilken et al., 2017). Finally, as the technology aims to have a full body picture of the person on the screen and does not allow real time interaction via the front camera, the presence of technology is very significant (i.e., low spatial presence: Hilken et al. (2017)). Therefore, although positioned as AR-based m-commerce tools, current applications offer reduced AR experience (Hilken et al., 2017; Poushneh and Vasquez-Parraga,

2017b), which might explain to some extent the reticence of the consumers to use them for shopping.

According to the respondents, the problem was not that the photo of the garment was small or static but that the technology itself could not translate the specificities of the textile and the human body. The respondents named videos, many photos of human models from every angle, and 360° representations of products as preferable formats for garment presentation. Moreover, the focus of the application was the virtual reality fitting, but the respondents desired more informativeness (Poushneh and Vasquez-Parraga, 2017b) and shopping-oriented functionalities, such as details on the fabric, price, and delivery. The initial excitement about the new technology changed to boredom and frustration with a slow application.

Interestingly, the older consumers (33 and older) voiced concerns related to the privacy policies and security of the uploaded photos. Poushneh (2018) also shows that there is an important trade-off between the quality of the AR features and the access to personal data. It is suggested that the possibility to control the access to personal information will improve the appreciation of the AR-based application (Poushneh, 2018).

As an aftermath, it seems that the existing technology and the small screen of mobile devices are not yet or not at all adapted for a full AR experience of all types of garments. Furthermore, other functions of the mobile shopping platforms, such free delivery and return, secure payment, social network (i.e., reviews) often prevail the function of a virtual fitting.

Table 9
Illustrative Excerpts and Main Conclusions of Study 2

Categories	Conclusions	Excerpts
Perceived convenience of examining the fit of the product	The unrealistic representation did not help examine product fit.	"It is just a flat image on a flat image. Apart from my face, it does not give me any idea whether it fits me or not."

Augmented reality experience and quality	The user cannot move and perform movements to adjust the garment to her body (*simulated physical control*)	"The garment is just a static photo on the static photo of my body. The dress is not well positioned on the photo. It is not well "aligned" with my body, and I can do nothing about that. I cannot grab it or move it."
	The app allows to select pictures of realistic backgrounds (indoor or outdoor) but does not show in real time the background of the user's own home *(environmental embedding)*	"Within the app you can adapt your body pose and make the surroundings realistic, but the dress does not look real." "It is not a mirror: your picture is quite static and the background is not real. Besides, the fact that the background can "move" is not enough to make it look real"
	The app does not offer an authentic situated experience in which the virtual content is located in the physical reality of the user and available for interactions *(spatial presence)*	"It works when I "try on" make up or small accessories, but for a dress it looks like I am small Barbie doll on the screen." "It is not realistic. It is not really me on that screen: I just see a bad picture of myself with a huge chest and short legs, and a photo of a garment that does not move* above my disproportionate body. It is not a like in a mirror."
Perceived enjoyment	At first, the application was enjoyable, but it soon became frustrating.	"It is a cool idea to be able to see the garment on yourself, but there is no movement, and it is just too artificial."*
Perceived ease of use	Use of the technology did not present any difficulties, but the virtual reality application often slowed down and crashed smartphones.	"It is just another app, quite easy to use." "It feels like a Snapchat. Unfortunately, it crashed on my phone after 3 minutes of use."
Perceived usefulness	The application was perceived as not useful because it lengthened the buying process and did not help to choose good garments.	"It might give only a first idea of what might be a good dress, but then you still need to inspect it on a better website or in person." "Overall, just for fun, no utility."
Data privacy and security	The respondents were uncomfortable uploading one's image on the app because of the security doubts.	"To make it look somewhat realistic I have to upload my picture in underwear…I am not sure that this app will protect my data, as it is something really personal"
Time efficiency	The time and effort needed to use the VTO outweighed the potential advantages.	"I don't want to be like my teenager, the whole day on the phone: finally, it is easier to try a garment in store or just order it

		and in case it does not fit return it."
Distribution over online fit evaluation	Respondents prioritized free delivery and return over virtual fitting.	"No, you can never know how the dress will fit based on the picture, you have to put it on. So, I prefer ** site: even if I cannot always have all the angles of the dress on the model, I take two sizes to try and I only leave something that I really like and that has a great fit."
Experience in online shopping	Respondents who accumulated experience in online shopping do not feel the need of a VTO.	"I have been buying clothes online for years now. I know my brands, I know my shapes, and my sizes. I do not need any virtual tool. It might be fun for a funny experiment, change of style, but for everyday shopping, I can tell better what fits me. But I do appreciate good photos, detailed information, and reviews."
Different physical control for different product types	Such tool seems to be more applicable to accessories rather than clothes.	"I would try a necklace or glasses on me, because the fit the same on every person, but how will see the fit of a dress on my back in an app?"
Attitudes toward the shopping technology	The idea of the application was exciting, but the actual performance was disappointing.	"It was a nice experience, but I do not think that I am going to use this app in the future."
Purchase intention	The respondents wanted to try on dresses before making decisions or to have more information on products and potential purchase options.	"I could not by on this application. If the app is designed to purchase online and avoid going to the store, it is clearly a failure… Because you do not know what size to choose with this app, so you cannot buy at all. You have to go to the store to check the size."** "I am not sure. I have to have another look at the dress before I choose."

Notes: * The AR-based VTO allowed consumers to move and create pictures of garments on photos or videos of themselves, but the technological solution did not allow visualization of how garments moved. ** Due to the complexity of the technology, the VTO had to be downloaded separately, and while it allowed access to certain apparel websites, the focus of the technology was the visualization of products in AR, not the overall shopping experience.

5. Discussion

5.1. Theoretical Implications

This research offers three important contributions. First, it contributes to the growing field of online product presentation in e-commerce (e.g., Beck and Crié, 2018; Yoo and Kim, 2014) and enriches the scant literature on the impacts of human models in product presentations (e.g., Berg, 2015; Kim et al., 2009; St-Onge et al., 2017). A key finding of this study is that human models function as important factor in realistic presentations of online products and stimulate better attitudes toward shopping technologies and higher purchase intention than AR-based VTO tools.

Second, whereas the advertising literature has offered varying results on the necessity to match human models and consumers (e.g., Chevalier and Lichtlé, 2012; Keh et al., 2016; Smeesters et al., 2010), the present research shows that in the m-commerce setting, a fit with the presenting human model (in terms of ethnicity and body shape) is likely to foster positive attitudes toward shopping technologies and higher purchase intention. Indeed, as shown by St-Onge et al. (2017), while thin models in the e-commerce setting are seen as more attractive and while the perceived attractiveness is directly related to the purchase intention, this relationship is negatively mediated by the actual size of the consumers. Thus, our results contribute to the marketing literature as the impact of the congruence between the consumer and the model in m-retailing has only been studied in the context of *virtual* models (Merle et al., 2012; Shim and Lee, 2011). Overall, this research offers a better understanding of the role of human models in online commerce product presentation. The e-commerce literature first identified the functional role of e-retailing human models: providing more accurate visualizations of products than flat presentations (Kim et al., 2009). Consistent with previous results showing that perceived congruence with virtual models creates a better experience on websites (Merle et al., 2012; Shim and Lee, 2011), the present research confirms the potential identification role of e-retailing human models. The results show that consumers identify themselves with the pictured

human models based on their ethnicity and, only in some cases, their body size.

Third, this research offers interesting conclusions on the relative impacts of a traditional m-commerce interface with pictures of human models and an AR-based VTO tool. Previous works compared AR-based VTO tools with traditional presentation formats on webstores (e.g., Beck and Crié, 2018; Yim et al., 2017) but did not study the impacts of the presenting human models on the webstore. Contrary to their findings that showed that AR-based VTO tools have more benefits than traditional presentation formats, we find that novel AR-based tools are not yet sufficiently technologically advanced to offer consumers enjoyable, helpful experiences and increase the effectiveness of online shopping. Traditional product presentation involving human models, therefore, remains preferable. However, as discussed, the success of this traditional presentation relies on the similarities in the appearance of the human model and the consumer.

Furthermore, this research contributes to the analysis of VTO tools' hedonic and utilitarian benefits and to the debate on their relative importance (e.g., Javornik et al., 2016; Rese et al., 2017). The hedonic benefits of VTO tools are presumed to be strong (e.g., Javornik et al., 2016; Kim and Forsythe, 2008) and are sometimes more important and more valued than the utilitarian and transactional aspects of AR applications (e.g., Scholz and Duffy, 2018). AR-based VTO tools are even supposed to result in higher enjoyment than traditional product presentations on webstores (Yim et al., 2017). Surprisingly, contrary to these results, we show that VTO can sometimes create less enjoyable shopping experiences than traditional product presentation. Moreover, while AR-based VTO tools have been proven to facilitate product's evaluation (Javornik et al., 2016) and generate greater usefulness than traditional presentations on webstores (Yim et al., 2017), we show in this research that the perceived utilitarian benefits of AR-based VTO tools are not always obvious. This result is consistent with previous findings that showed that consumers are sometimes skeptical about the information on garments' fit provided by VTO tools (e.g., Kim and Forsythe, 2008). AR-based VTO tools are indeed very

different from an actual try-on of a garment: they can provide consumers with information related to the garment's attributes (i.e., size or color) but they often fail to help consumers evaluate the garment's fit with themselves (Baytar et al., 2016). Possible explanations for the lack of hedonic and utilitarian benefits of AR-based VTO tools have been brought in this research. Thus, technological limitations of current AR-based VTO tools and poor augmentation quality (e.g., Poushneh, 2018; Poushneh and Vasquez-Parraga, 2017b) could explain their lack of success. At last, it is important to determine the relative importance of hedonic and utilitarian benefits in AR-based VTO tools adoption or evaluation (e.g., Javornik et al., 2016; Rese et al., 2017). Our research brings interesting insights in that matter and shows that perceived enjoyment from an m-commerce interface with an AR-based VTO has a stronger effect on consumer attitudes than utilitarian perceptions.

5.2. Managerial Implications

Our findings offer several managerial insights for apparel e-retailers. First, companies should sufficiently invest in AR-based VTO tools to make sure the experience offered is perceived as both enjoyable and practical (i.e., useful and convenient for examining products before purchasing). If such a tool is not available, retailers should focus on the presentation of garments on human models and make sure that the featured human models correspond to the appearance of the targeted audience. Adapting human models to consumers' appearance is recommended. For example, the Japanese retailer Uniqlo uses non-Asian and more full-bodied human models in the United States because approximately 73.9 of North Americans are overweight (Walpole et al., 2012) and only about 6% have Asian ethnicity[3]. Furthermore, to increase the potential to fit consumers' appearances, it is recommended that retailers feature different human models wearing the same garment (as employed by PrettyLittleThing[4]). These measures will lead both to better attitudes toward brands' m-commerce interfaces and to higher purchase intention for the featured garments.

Moreover, from our qualitative study it becomes clear that consumers take into great consideration the informativeness, functionality, the terms of sale on the m-commerce application, and the quality of the offline service. Therefore, companies should evaluate whether their target consumers are prone to use a novel tool. For instance, the virtual try rooms that are being developed by Gap[5] stimulate a cold welcome rather than enthusiasm among the consumers.

5.3. Limitations and Future Research

This research has limitations that could be overcome in future studies. In the first hand, there are some limitations that are related to the experimental design. The VTO condition in the quantitative experiment was introduced by a video, made by the developers of the app. It is possible that the way it was made impacted the perceptions and attitudes of the respondents. Furthermore, no sophisticated AR-based VTO is currently available for downloading on the mobile application platforms. Thus, the application used for the qualitative follow-up study allowed only limited AR experience, which might explain to a large extent why the respondents did not find it particularly convenient or useful. Nevertheless, this study offers valuable insights into the needs and perception of mobile shoppers and provides directions for further development of AR-based tool in m-retailing. Further research should address these technological aspects, once the next-stage AR-enhanced mobile applications are available, and compare the full AR-based VTO tool to the interactive functions of the traditional m-commerce applications, such as zooming, rotating, having a 360° view, or a video.

In the second hand, other limitations are related to the research sample and framework. Thus, it would be interesting to investigate the impacts of e-commerce human models on consumers from different backgrounds and cultures. We defined ethnicity as race in this research. Race is a social construction rather than a strictly biological classification (Jones, 2010).

The United States adopt strict racial classifications with a dichotomy between White and Black whereas others countries such as Brazil (where mix-raced individuals are very represented) adopt fluid classifications that run along a continuum between Black and White (Jones, 2010). As the literature usually recognizes the existence of the Asian, Black, and White races and studies their impact in consumer research (e.g., D'Alessandro and Chitty, 2011; Jones, 2010; Keh et al., 2016), this research has adopted this classification. However, this approach of racial classification is narrow and has limitations. Therefore, it seems important to include mix-raced identities and specific ethnicities in future research (e.g., Latin ethnicities, South Asian ethnicities, etc.). Moreover, the proposed framework could incorporate the interactions among human models' ethnicity, body size, and other physical and situational attributes, such as attractiveness (e.g., Buunk and Dijkstra, 2011), age (e.g., Chevalier and Lichtlé, 2012), and poses in pictures (e.g., Cazzato et al., 2012; Khakimdjanova and Park, 2005). Other variables might also be used to enrich the conceptual framework. For instance, following the suggestions of Hilken et al. (2018), the framework could incorporate variable understudied variables in the AR research, such as information gathering or decision confidence. Also, psychographic variable relevant to the decision making in the context should be taken into account (e.g., mental imagery ability and need for touch (Hilken et al., 2017)).

Finally, the framework could focus more specifically on the impact of AR-based VTO tools on consumers' evaluations of the garment's size and fit (e.g., Baytar et al., 2016; Kim, 2015; Shin and Baytar, 2014). AR-based VTO tools are considered as less effective than an actual try-on to evaluate the garment's fit (Baytar et al., 2016). Further research should focus on that topic and determine if an AR-based VTO tool is more effective than a picture of a human model to evaluate a garment's fit. This might discuss with the surprising results of Shin and Baytar (2014) who showed that a picture of an idealized human model rises less concerns about the garment's fit than a 3D avatar built with the consumer's own measurements.

Footnotes

[1] https://www.emarketer.com/Chart/Retail-Ecommerce-Sales-Worldwide-2017-2021-trillions-change-of-total-retail-sales/219928

[2] https://www.emarketer.com/Report/Worldwide-Retail-Ecommerce-Sales-eMarketers-Updated-Forecast-New-Mcommerce-Estimates-20162021/2002182

[3] http://www.pewresearch.org/fact-tank/2017/09/08/key-facts-about-asian-americans/

[4] https://www.cosmopolitan.com/uk/fashion/style/a24876115/pretty-little-thing-different-size-models/

[5] https://www.dailymail.co.uk/sciencetech/article-4089338/Gap-reveals-new-app-lets-virtually-try-clothes-home.html

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Appendices

Appendix 1
Example of visual used in the experiment manipulation

Figure A1.1: Asian human model thin (left) and curvy (right)

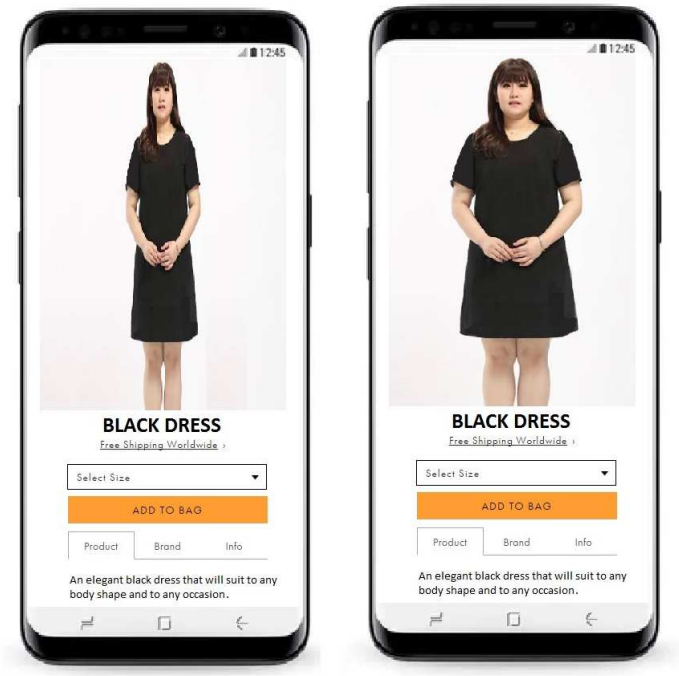


Figure A1.2: Black human model thin (left) and curvy (right)

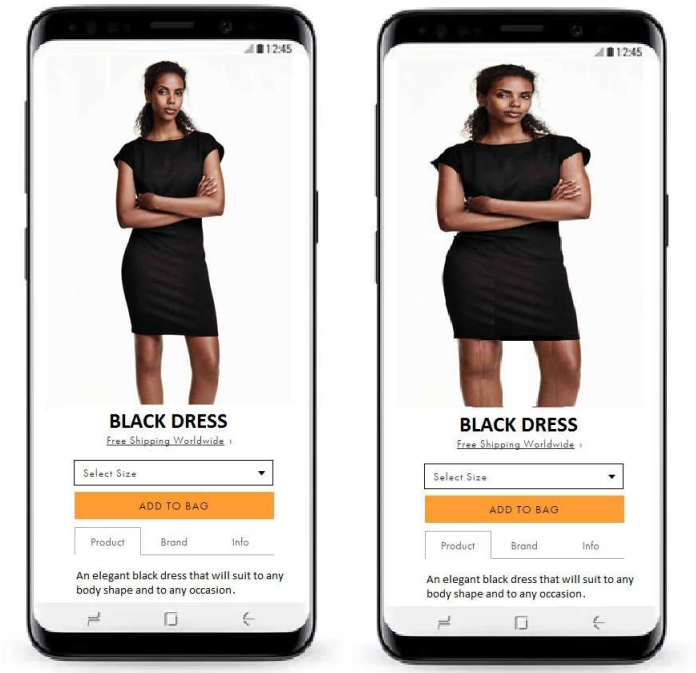
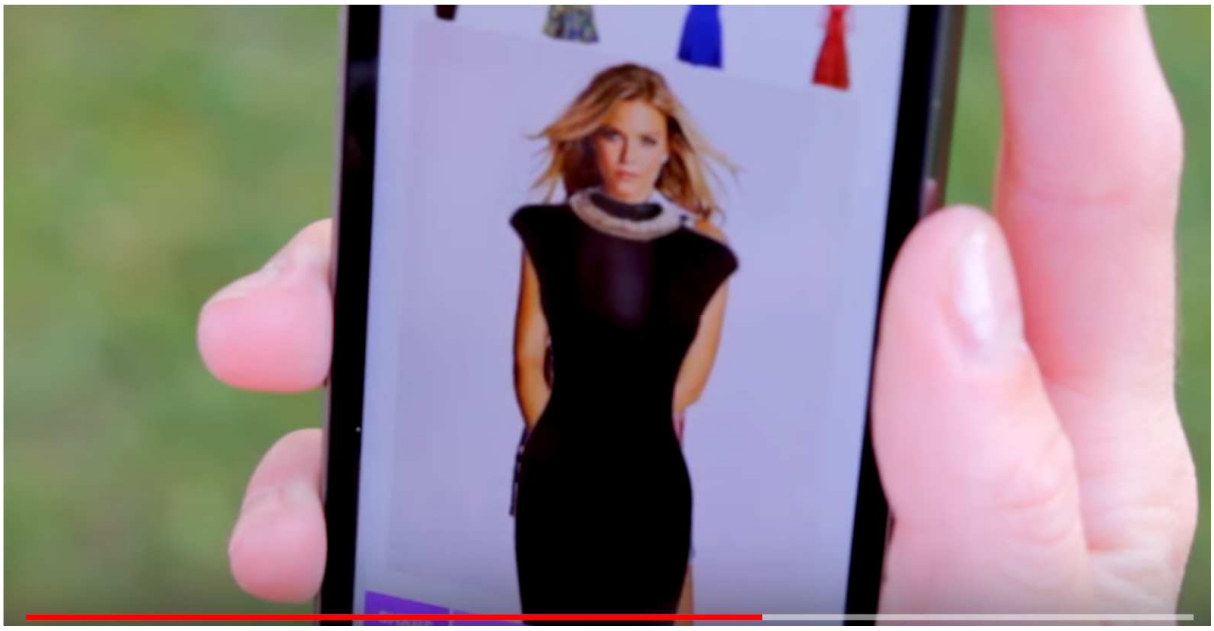


Figure A1.3: Screenshot of the shown video on the AR-based VTO mobile application



Appendix 2

Operationalization of constructs and measurement items.

Construct	Measurement items
Purchase intention: adapted from Chandran and Morwitz (2005)	PI1. It is very likely that I would purchase this outfit. PI2. The probability that I would consider buying the product is very high. PI3. If I were going to buy clothes, I would very probably buy this dress.
Attitude toward the shopping technology: adapted from Chen et al. (2002)	AT1. This is a good shopping technology. AT2. Assuming I had access to the shopping technology, I intend to use it. AT3. Given that I had access to the shopping technology, I predict that I would use it.
Convenience to examine the product: adapted from Childers et al. (2001)	CNV 1. Use of this shopping technology will offer knowledge of a product similar to that available from a personal examination. CNV 2. Use of this shopping technology would allow me to form an impression about a product similar to that from up-close examination. CNV 3. Use of this shopping technology would allow me to judge a products' quality as accurately as an in-person appraisal of the product.
Perceived enjoyment from the shopping experience: adapted from Childers et al. (2001)	ENJ1. Shopping with this shopping technology would be fun for its own sake. ENJ2. Shopping with this shopping technology would involve me in the shopping process. ENJ3. Shopping with this shopping technology would be exciting. ENJ4. Shopping with this shopping technology would be enjoyable.
Perceived ease of use of the shopping technology: adapted from Davis (1989)	EOU1. This shopping technology would be clear and understandable. EOU2. This shopping technology would not require a lot of mental effort. EOU3. This shopping technology would be easy to use.
Perceived usefulness of the shopping technology: adapted from David (1989)	US1. This shopping technology would improve my shopping productivity. US2. This shopping technology would enhance my effectiveness in shopping. US3. This shopping technology would be useful in buying what I want.
Evaluation of the dress: adapted from Bower and Landreth (2001)	DR1. Good/bad. DR2. Likable/dislikable. DR3. Of high quality/of poor quality. DR4. Worth buying/not worth buying.
Involvement in online retailing shopping: adapted from Bower and Landreth (2001)	Online retailing shopping is: INV1. Unimportant / important to me INV2. Of no concern / of concern to me INV3. Irrelevant / relevant to me INV4. Doesn't matter // matters to me
Experience with shopping technologies: adapted from Griffith and Chen (2004)	In regard with online shopping technologies you: EX1. Know very little about / know very much about EX2. Inexperienced / experienced EX3. Uninformed / informed EX4. Novice buyer / expert buyer
Attention check: adapted from Oppenheimer et al. (2009)	To demonstrate attention to the instructions you should ignore the actual question and instead tick the "read the instructions" box.

Appendix 3

Constructs' validity and reliability.

	Item	Component Loading***	Variance Explained	Reliability*
Purchase intention	PI1	0.949	90.28%	α= 0.951
	PI2	0.951		AVE= 0.903
	PI3	0.951		CR=0.965
Attitude toward the shopping technology	AT1	0.900	91.14%	α=0.906
	AT2	0.918		AVE= 0.842
	AT3	0.936		CR=0.941
Perceived convenience	CNV1	0.911		α= 0.900
	CNV 2	0.933		AVE= 0.834
	CNV 3	0.896		CR=0.937
Perceived enjoyment	ENJ1	0.942	66.96%	α=0.822
	ENJ2	0.961		AVE= 0.917
	ENJ3	0.998		CR=0.962
	ENJ4	0.929		
Perceived ease of use	EOU1	0.892	65.53%	α= 0.905
	EOU 2	0.934		AVE= 0.840
	EOU 3	0.924		CR=0.940
Perceived usefulness	US1	0.862	74%	α= 0.831
	US2	0.886		AVE=0.901
	US3	0.850		CR=0.900
Evaluation of the dress	DR1	0.949	87%	α= 0.951
	DR2	0.950		AVE=0.871
	DR3	0.900		CR= 0.938
	DR4	0.935		
Involvement with online shopping	INV1	0.951	71.44%	α= 0.950
	INV2	0.948		AVE=0.829
	INV3	0.899		CR=0.919
	INV4	0.841		
Experience with online clothes shopping	EX1	0.943	71.23%	α= 0.924
	EX2	0.976		AVE=0.929
	EX3	0.988		CR=0.967
	EX4	0.949		

* α: Alpha Cronbach; AVE: Average Variance Extracted; CR: Composite Reliability

Appendix 4

Correlation matrix

	Purchase intention	Attitude toward the shopping technology	P. convenience	P. enjoyment	P. ease of use	P. usefulness	Evaluation of the dress	Involvement with online shopping	Experience with online clothes shopping
Purchase intention	1	.704**	.600**	.489**	.415**	.549**	212*	.348**	.181**
Attitude toward the shopping technology	.704**	1	.478**	.704**	.629**	.691**	.164*	.341**	.244**
Perceived convenience	.600**	.478**	1	.567**	.241**	.373***	.112*	.346**	.108*
Perceived enjoyment	.489**	.560**	.567**	1	.372**	.406**	.205*	.279**	.262**
Perceived ease of use	.415**	.629**	.241**	.372**	1	.715**	.270*	.310**	.420**
Perceived usefulness	.549**	.691**	.373**	.406**	.715**	1	.198*	.330**	.284**
Evaluation of the dress	212*	.164*	.112*	.205*	.270*	.198*	1	.103*	.142*
Involvement with online shopping	.348**	.341**	.346**	.279**	.310**	.330**	.103*	1	.282**
Experience with online clothes shopping	.181**	.244**	.108*	.262**	.420**	.284**	.142*	.282**	1

**. Correlation is significant at the 0.01 level
*. Correlation is significant at the 0.05 level

Appendix 5

Post-Hoc between-group results

Dependent variable	Condition (A)	Mean (A)	Condition (B)	Mean (B)	Significance of the means difference (A vs. B)
Perceived convenience of examining the product	**VTO**	**4.81 (1.44)**	**Full Fit**	**5.37 (1.11)**	**p-value<.001**
			Ethnic Fit	4.92 (1.54)	ns
			Body Fit	4.93 (1.64)	ns
			No Fit	**4.41 (1.71)**	**p-value<.010**
	Full Fit	**5.37 (1.11)**	Ethnic Fit	4.92 (1.54)	ns
			Body Fit	4.93 (1.64)	ns
			No Fit	**4.41 (1.71)**	**p-value<.001**
	Ethnic Fit	4.92 (1.54)	Body Fit	4.93 (1.64)	ns
			No Fit	4.41 (1.71)	ns
	Body Fit	4.41 (1.71)	No Fit	4.41 (1.71)	ns
Perceived enjoyment	**VTO**	**3.28 (1.42)**	**Full Fit**	**5.41 (1.21)**	**p-value<.001**
			Ethnic Fit	**5.5 (0.90)**	**p-value<.001**
			Body Fit	**5.17 (1.54)**	**p-value<.001**
			No Fit	**5.17 (1.56)**	**p-value<.001**
	Full Fit	5.41 (1.21)	Ethnic Fit	5.5 (0.90)	ns
			Body Fit	5.17 (1.54)	ns
			No Fit	5.17 (1.56)	ns
	Ethnic Fit	5.5 (0.90)	Body Fit	5.17 (1.54)	ns
			No Fit	5.17 (1.56)	ns
	Body Fit	5.17 (1.54)	No Fit	5.17 (1.56)	ns
Perceived ease of use	VTO	5.26 (1.47)	Full Fit	5.55 (1.28)	ns
			Ethnic Fit	5.61 (0.95)	ns
			Body Fit	5.64 (1.29)	ns
			No Fit	5.53 (1.24)	ns
	Full Fit	5.55 (1.28)	Ethnic Fit	5.61 (0.95)	ns
			Body Fit	5.64 (1.29)	ns
			No Fit	5.53 (1.24)	ns
	Ethnic Fit	5.61 (0.95)	Body Fit	5.64 (1.29)	ns
			No Fit	5.53 (1.24)	ns
	Body Fit	5.64 (1.29)	No Fit	5.53 (1.24)	ns
Perceived usefulness	**VTO**	**4.85 (1.50)**	Full Fit	5.22 (1.13)	ns
			Ethnic Fit	**5.62 (0.96)**	**p-value<.050**
			Body Fit	5.34 (1.40)	ns
			No Fit	5.10 (1.41)	ns
	Full Fit	5.22 (1.13)	Ethnic Fit	5.62 (0.96)	ns
			Body Fit	5.34 (1.40)	ns
			No Fit	5.10 (1.41)	ns
	Ethnic Fit	5.62 (0.96)	Body Fit	5.34 (1.40)	ns
			No Fit	5.10 (1.41)	ns
	Body Fit	5.34 (1.40)	No Fit	5.10 (1.41)	ns
Attitudes toward the shopping technology	VTO	5.12 (1.52)	Full Fit	5.73 (1.18)	ns
			Ethnic Fit	5.60 (1.08)	ns
			Body Fit	5.26 (1.45)	ns
			No Fit	5.38 (1.45)	ns
	Full Fit	5.73 (1.18)	Ethnic Fit	5.60 (1.08)	ns
			Body Fit	5.26 (1.45)	ns
			No Fit	5.38 (1.45)	ns
	Ethnic Fit	5.60 (1.08)	Body Fit	5.26 (1.45)	ns
			No Fit	5.38 (1.45)	ns
	Body Fit	5.26 (1.45)	No Fit	5.38 (1.45)	ns
Purchase intention	VTO	4.84 (1.58)	Full Fit	5.42 (1.44)	ns
			Ethnic Fit	5.28 (1.15)	ns
			Body Fit	4.90 (1.86)	ns
			No Fit	4.65 (1.91)	ns
	Full Fit	**5.42 (1.44)**	Ethnic Fit	5.28 (1.15)	ns
			Body Fit	4.90 (1.86)	ns
			No Fit	**4.65 (1.91)**	**p-value<.050**

	Ethnic Fit	5.28 (1.15)	Body Fit	4.90 (1.86)	ns
			No Fit	4.65 (1.91)	ns
	Body Fit	4.90 (1.86)	No Fit	4.65 (1.91)	ns